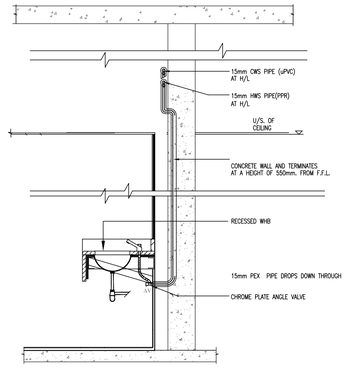


NOTES:

1.PUMP SHALL BE END SUCTION BACK PULLOUT CENTRIFUGAL TYPE
2.UNIT SHALL OPERATE ON 415V, 3PH, 50HZ ELECTRICAL POWER SUPPLY CHARACTERISTICS.
3.STATED PUMP HEAD IS INDICATIVE ONLY. CONTRACTOR SHALL COMPUTE ACTUAL HEAD BASED OON SITE CONDITIONS AND SHOP DRAWINGS. SIZE PUMP AND DRIVE MOTOR SUITABLY
4.ALL PARTS OF PUMP IN CONTACT WITH WATER SHALL BE OF HIGH GRADE STAINLESS STEEL (AISI 316).
5.ALL ITEMS USED IN PUMP ASSEMBLY SHALL BE FROM MANUFACTURERS AS APPROVED FOR THE PROJECT.
6.ALL CAST IRON & STEEL PARTS USED IN PACKAGING PUMP SHALL BE APPLIED INTERNALLY AND EXTERNALLY WITH EPOXY PAINT SYSTEM TO 200 MICRON DFT.

TYPICAL DETAIL FOR BOOSTER WATER PUMPSET



TYPICAL INSTALLATION OF HOT & COLD WATER SUPPLY FOR WHB UNITS

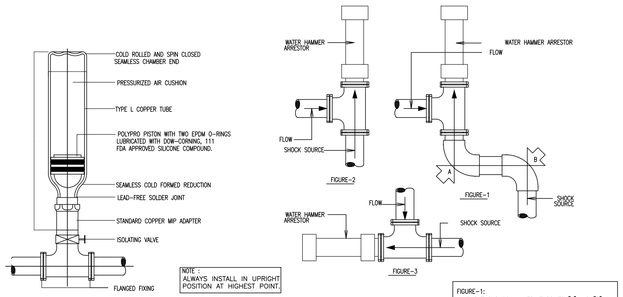


FIGURE-1: ALLOWS TWO SHOCK WAVES AT POINTS "A" AND "B" WHICH COULD RESULT IN SHAKE AND RATTLE OF THE PIPES WHEN THE VALVE IS CLOSED.
FIGURE-2 & 3: ILLUSTRATE PROPER POSITIONING WHERE THE SHOCK WOULD BE ABSORBED WITH 100% EFFECTIVNESS.

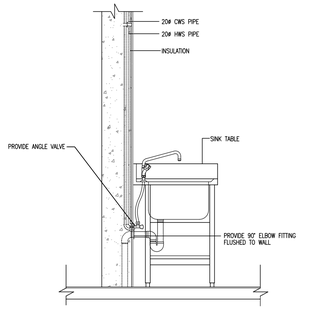
NOTE:-
WATER HAMMER ARRESTORS SHOULD ALWAYS BE PLACED AS NEAR TO THE SOURCE OF SHOCK AS POSSIBLE – ONE EACH ON THE HOT AND COLD WATER LINES.

TYPICAL WATER HAMMER ARRESTER

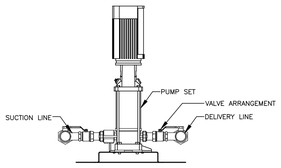
SL. NO.	SANITARY FIXTURE	HOT mm	COLD mm
1	WATER CLOSET (WC) WITH CISTERN	–	15
2	WATER CLOSET (WC) WITH FLUSH VALVE	–	40
3	WASH BASIN	15	15
4	ABLUTION HOSE	–	15
5	BATHTUB	20	20
6	SHOWER	20	20
7	BIDET	15	15
8	KITCHEN SINK	15	15
9	WASHING MACHINE	–	15
10	DISH WASHER	15	15
11	URINAL	–	15
12	URINAL WITH FLUSH VALVE	–	32
13	CLEANERS SINK	–	15

EQUIVALENT PIPE DIAMETERS

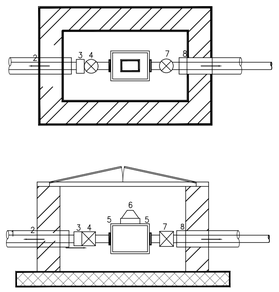
NOMINAL PIPE DIA(mm)	COPPER PIPE DIA(mm)	POLYBUTYLENE (mm)	PEX (mm)	POLYPROPYLENE (PPR) (mm)
15	15	20	20	25
20	22	25	25	32
25	28	32	32	40
32	35	40	N/A	50
40	42	50	N/A	63
50	54	63	N/A	75
65	67	75	N/A	90
80	76	90	N/A	110
100	108	100	N/A	N/A
125	133	125	N/A	N/A
150	159	150	N/A	N/A



TYPICAL INSTALLATION OF HOT & COLD WATER SUPPLY AND DRAINAGE PIPE FOR SINK UNITS



TYPICAL DETAIL OF TRANSFER PUMP SUBMERSIBLE TYPE

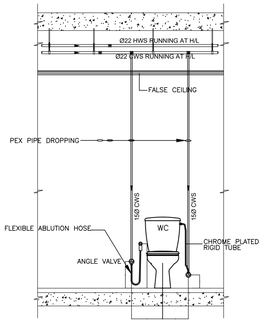


LEGEND

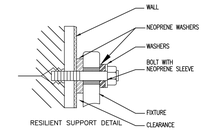
1. POLYETHENE
2. PVC DUCT
3. G.M.P.F ADAPTER
4. GATE VALVE
5. NOZZLES
6. WATER METER
7. GATE VALVE

TYPICAL DETAIL OF WATER METER FIXING

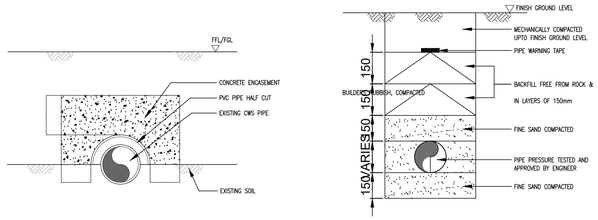
Fixture type	Maximum flow rate
Shower Head	≤ 5.6 L/min
Hand Wash Basin	≤ 4.2 L/min
Kitchen Sink	≤ 5 L/min
Dual Flush Toilet	≤ 4.5 L full Flush, ≤ 3 L part flush
Urinal	≤ 1.68 L in non-public facilities ≤ 0.7 L per flush or waterless in public facilities



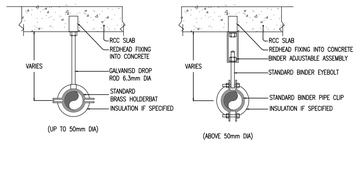
TYPICAL DETAIL OF WATER CLOSET



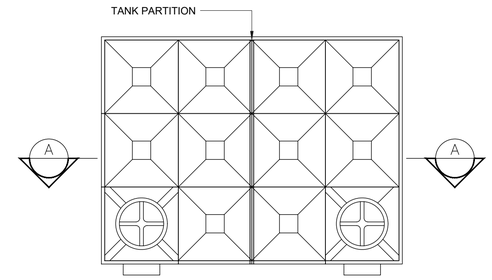
TYPICAL BATHROOM FIXTURE ISOLATION DETAIL



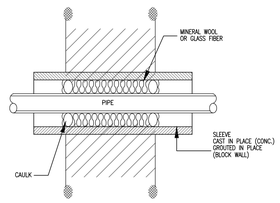
TYPICAL BEDDING DETAILS OF UNDER GROUND WATER PIPES



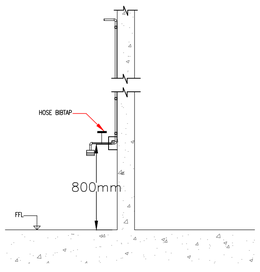
STANDARD SINGLE PIPE LINE HANGERS FOR WATER SERVICES



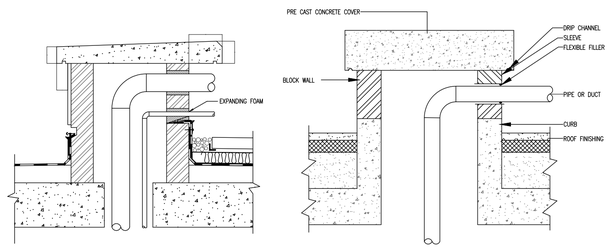
TANK PLAN WITH PARTITION



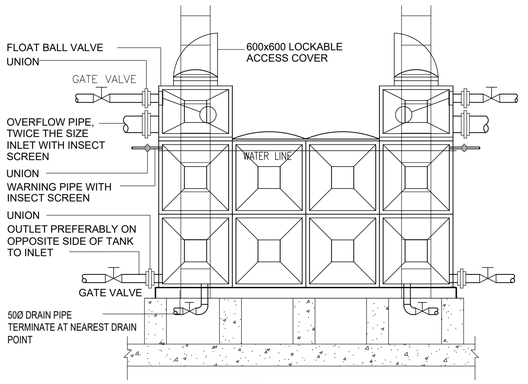
WALL PENETRATION



DETAIL OF HOSE BIBTAP



TYPICAL PIPE/DUCT WORK AT ROOF FOR MULTIPLE SERVICES



SECTION AA

TYPICAL WATER TANK SECTION DETAILS

DATE	DESCRIPTION	BY

CONSULTANT :

EMSQUARE
ENGINEERING CONSULTANTS
ARCHITECTS-CIVIL ENGINEERS-INTERIORS
P.O. Box 181903, Tel: +971 4 2729009, Fax: +971 4 2729334
Email: emsquare@emirates.net.ae

OWNER :

G F S REAL ESTATE L.L.C

JOB TITLE :

PROPOSED RESIDENTIAL BUILDING (B+G+6+ROOF)

PLOT NO. / PARCEL ID	BLOCK / COMMUNITY
5310301	SAIH SHUAIB 2

DRAWING TITLE :

TYPICAL DETAILS
WATER SUPPLY LAYOUT

ARCHITECT :	STRUCTURAL ENGR. :	DRAWN :
LAYLA	-	SATHEESH
SCALE :	DATE :	CHECKED BY :
NTS @ A1	AUGUST2025	YASEEN
JOB NO :	DRAWING NO. :	REVISION NO. :
----	WS-11	--